Final project quantum computing report

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***Abstract*—The goal of the project was to make a program that can transform a particular logic file into the realizable quantum circuit.**

I. Introduction

At the start I made a program in C language but then after having some difficulties due to the architecture of the language I redid the program on python. The first thing I made was an algorithm that allows me to read pla file. Due to the formatting it wasn't difficult to make since comments lines could be skipped if the first character of the line is hash symbol. Saving the input and output sizes was also done pretty quickly but from what I noticed it was not necessarily done in the same format in every .pla file so I had to implement an algorithm to check the order in the code. The first run of my program is in order to check how many lines it needs to skip to collect input output data matrices and in order to know the amount of variables in the output and input. My first idea was to use the given amount of input and output variables to dynamically allocate the arrays and then fill them with the data from the .pla file. I had to reject that idea due to various circumstances such as time restriction and it wouldn't work with integers and Boolean arrays due to the fact that there was possible a '-' symbol in the arrays. So what I did is I saved them as a string and due to the fact that its possible to compare string types it ended up being a very convenient choice. Next run collects all essential data as well as saves the names of the variables. After that my first idea was to run a check of both input and output variables in order to find the rows where they differ. Since rows with no changes give us almost no information on what logical algorithm was used. Next step was to compare the differing rows and find out whether those rows have bits flipped and which bits were at 1 when that happened. This way I can define large multiple control Toffioli gates that could have been used. After finding those depending on amount of the bits that control the target I define them as multiple controlled toffoli gates. In case of two controlled in the Revlib there already was a suitable representation in logical circuit way. For the larger cases there is a different approach needed. I managed to get information on how to do it theoretically, however i was unable to make a code implementation of it due to the time constraint. Final part of the program is done so i can output the final resulting circuit in the .real format. The program output I made uses data it got from initial pla file then appropriately types it in the .real file you want to save. However, since .real format circuit does not have a section that shows the computational cost i had to output it differently.

II. Result

There a lot to improve upon in this work and as an author of the program i see a lot of problems in the code that i did not fixed. I didn't check if it will work for asymmetric cases where we can have garbage output and comparison of the input output won't work as a method cause theres not enough data to compare to. I researched on the asymmetric cases but i was unable to come up with a strategy that would allow me to reliably obtain the correct circuit as an answer.

December 5, 2019